IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Palanza Toledo, Martín Uriel
February 18, 2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies:
  - Data Engineering: SpaceX API, Web Scraping
  - Advanced Analysis: Geospatial visualization with Folium and relational analysis using SQL to intetify critical patterns
  - Predictive Modeling: Evaluation of machine learning algorithms (SMV, Decision Trees, Logistic Regression and KNN) optimized by Grid Search
- Summary of all results
  - Model Accuracy: An accuracy of 83.3% was achieved in the prediction of landings,
  - Decision tool: An interactive Dash app that allows simulating scenarios based on payload and launch site, facilitating strategic decision-making

Introduction

- Project background and context:
  - SpaceX has revolutionized the aerospace industry by reducing costs(from 165 millions dollars to 62 millons dollars) through the reuse of the first stage. However, each attempt to land the first stage involves logistical and financial risks.
- Problems you want to find answers
  - The challenge is to determine, before the launch,whether the stage recovery will be successful based on technical and environmental variables.
  - What determines if the first stage will land succesfully?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Rest API (spaceX): Web Scraping (Wikipedia).
- Perform data wrangling
  - Normalization to handle NaN values (impute median in payload mass); Transformation for categorical values using One-Hot-Encoding; Creation of new attribute 'class' to tag successful an failed launches.
- Perform exploratory data analysis (EDA) using visualization and SQL
  - SQL (SQLite): complex queries to identify tendencies.

Methodology

Executive Summary

- Perform interactive visual analytics using Folium and Plotly Dash
  - Geospatial visualization: using of Folium, to map proximity of launch sites to railroads, roads and coasts. Using of Dash app to simulate scenarios based on payload and launch site.
- Perform predictive analysis using classification models
  - Using of StandardScaler before to split dataset into training and test sets. Fitting of the following algorithms: Logistic Regression, SVM, Decision Trees, KNN. For each one GridSearchSV was used with 10-fold Cross-Validation.

Data Collection

- SpaceX API:
  - Data collection using get request to API.
  - Convert received JSON into Pandas DataFrame and fill missing values
  - Export to csv file
- Web Scraping:
  - Perform web scraping with BeautifulSoup from Wikipedia page from Falcon 9
  - Extract HTML table, parse and convert it into Pandas DataFrame

Data Collection – SpaceX API

- Data collection using get request to API.
- Filter only Falcon 9 launches.
- Extraction loop: for each row the script calls specific functions.
- Merge lists into a dict and transform received JSON into a Pandas DataFrame.
- Handle missing values by mean imputation.

[REQUEST] → [FILTER (Falcon 9)] → [EXTRACT (IDs)] → [LOOKUP (Details)] → [CLEAN (Mean Imputation)] → [DATAFRAME]

Completed SpaceX API calls notebook

Data Collection - Scraping

- Request: Obtaining HTML from Wikipedia by Request library.
- BeautifulSoup: Creation of "soup" objet to navigate the page's DOM.
- Localization of tables with class "wikitable".
- Extraction with "for" loop that iterates through each row (tr).
- Text extraction from each cell (td). Here we clean references (like [1], [2]) and handle rows without links.
- Storage in dict an transform into Pandas Dataframe.
- [WIKIPEDIA URL] → [BEAUTIFULSOUP] → [TABLE EXTRACTION] → [DATA CLEANING] → [PANDAS DATAFRAME]

Completed web scraping notebook

Data Wrangling

- Data load from csv.
- Identification of NaN values and fill with mean imputation.
- Categorizing 'Outcome' column to identify successful and failed landings.
- Labeling: Creation of 'class' column (target variable):
  - 1: Success (True Ocean, True RTLS, True ASDS).
  - 0: Fail (False Ocean, False RTLS, False ASDS, None Ocean, None ASDS).
- Export final dataset to .csv file.

[ORIGINAL CSV] → [TREATMENT OF NULLS] → [OUTCOMES CONVERSION] → [ONE-HOT-ENCODING] → [CREATE TARGET VARIABLE] → [FINAL DATASET]

Completed data wrangling notebook

EDA with Data Visualization

- Scatter plot Flight Number vs. Launch Site: used to observe if accumulated experience improves success rate in speceific Launch sites
- Scatter plot Payload Mass vs. Launch Site: used to determinate if exist a "critical load" that makes landing difficult at certain sites.
- Barplot Orbit vs. Success Rate: Fundamental to identify 'safest' orbits to the reuse of first stage.
- Lineplot Year vs. Average Success Rate: to demonstrate SpaceX's learning curve.

Completed EDA with Data Visualization Notebook

EDA with SQL

- Display the names of unique Launch Sites.
- Display the total payload mass carried by boosters launched by NASA (CRS).
- Display average payload mass carried by booster version F9 v1.1.
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload greater than 4000 but less than 6000.

EDA with SQL

- List the total number o successful and failure missions outcomes.
- List all the booster_versions that have carried the maximum payload mass
- List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.
- Rank the count of landing outcomes between the date 2010-06-04 and 2017-03-20, in descending order.

Completed EDA with SQL notebook

Build an Interactive Map with Folium

- Map objects created and added to a folium map:
  - Circle and marker for each Launch site
  - MarkerCluster to mark success/failed launches for each site on map.
  - PolyLine to represent and indicates distances between launch site VAFB SLC-4E an nearest coast, railroad and road.
- Why add those objects:
  - Marker cluster: to manage overlapping launches at the same site. Allows to see at a glance the total number of missions per base and identify geographical performance patterns,
  - Polylines: allows to represent distance between launch sites and places of interests.

Completed interactive map with Folium map.

Build a Dashboard with Plotly Dash

- Dash app allows to filtrate the dataset by two main controls:
  - Dropdown: to select launch site (or all launch sites).
  - RangeSlider: to select payload.
- Pie Chart:
  - Shows proportion of success and failures. Allow to compares the efficiency of each launch site. When "All" is selected, the graph shows which site contributes the most successes to SpaceX's overall program.
- Scatter Plot:
  - Shows relation between Payload Mass (horizontal axis) and Landing Result (vertical axis), the colors indicates booster versions. Allows to analyze if Payload and Booster Version influences the rocket's recovery capability.

Completed Plotly Dash script.

Predictive Analysis (Classification)

- Built and preprocessing:
  - Using of StandardScaler to normalize numerical variables, ensuring that no singles variables dominates over other. Trained models: Logistic Regression, SVM, Decision Trees, K-Nearest Neighbors.
- Optimization With GrisSearchCV
  - Using of GridSearch with Cross Validation to try combinations of parameters and select the optimal ones. 10-fold Cross-Validation scheme was used to ensure consistent model performance.
- Evaluation and selection of best model:
  - Models was evaluated using Accuracy and Confusion Matrix.

[INPUT] → [TRANSFORM] → [SPLIT] → [GRIDSEARCH LOOP] → [TEST] → [COMPARE]

Completed predictive analysis lab

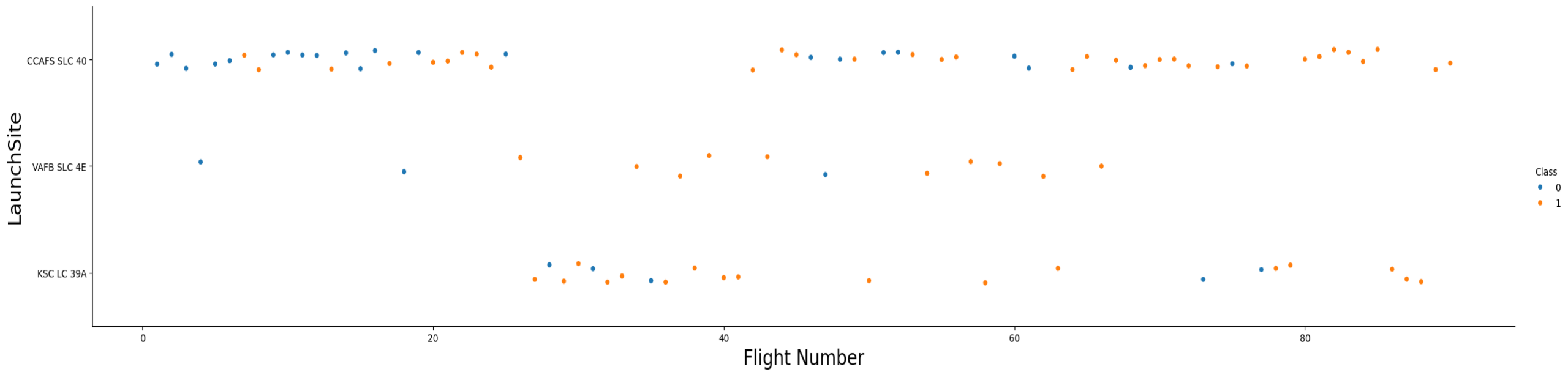
Results

- Exploratory data analysis results
  - As the number of flights increases, the density of successful landings increases at all sites, especially at KSC LC 39A.
  - The VAFB SLC 4E site does not handle heavy loads, while for very heavy loads at other sites, the success rate appears to be higher, suggesting that these are missions with more modern and optimized flight profiles.
  - The ES-L1, GEO, HEO, and SSO orbits have a 100% success rate, while the GTO orbit is usually more challenging.
  - The lineplot shows a clear upward trend since 2013, demonstrating that reuse technology has matured significantly over time.
- Predictive analysis results
  - Surprisingly, all four models achieved an identical accuracy of 83.33% on the test data. Tie-breaking criterion: In cases of equal accuracy, the model with the greatest interpretability, such as Logistic Regression or Decision Tree, is usually preferred.

Section 2

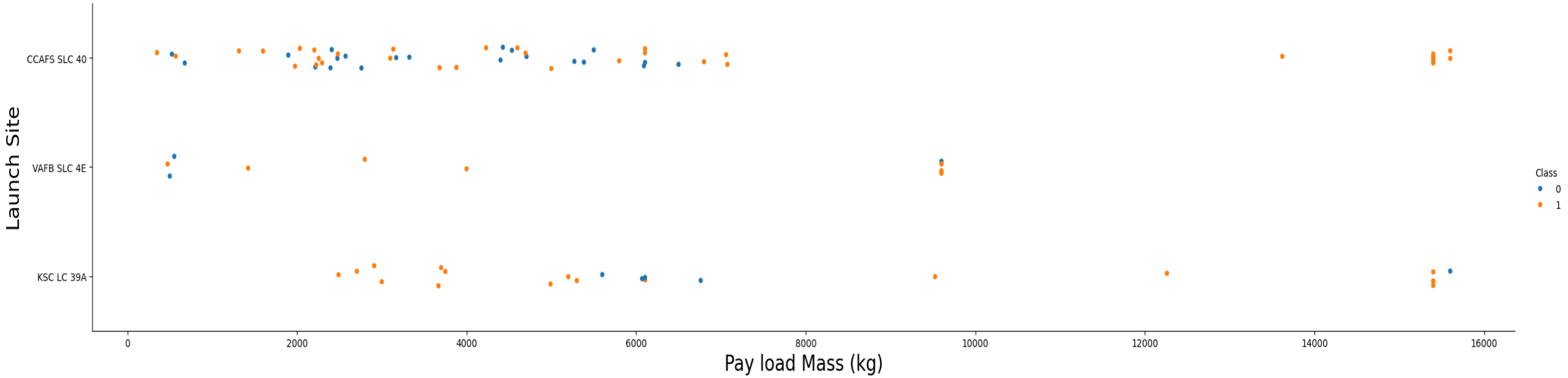
Insights drawn from EDA

Flight Number vs. Launch Site



- It can be seen that as increases 'Flight Number', increases density of successful landings, mainly at KSC LC 39A.

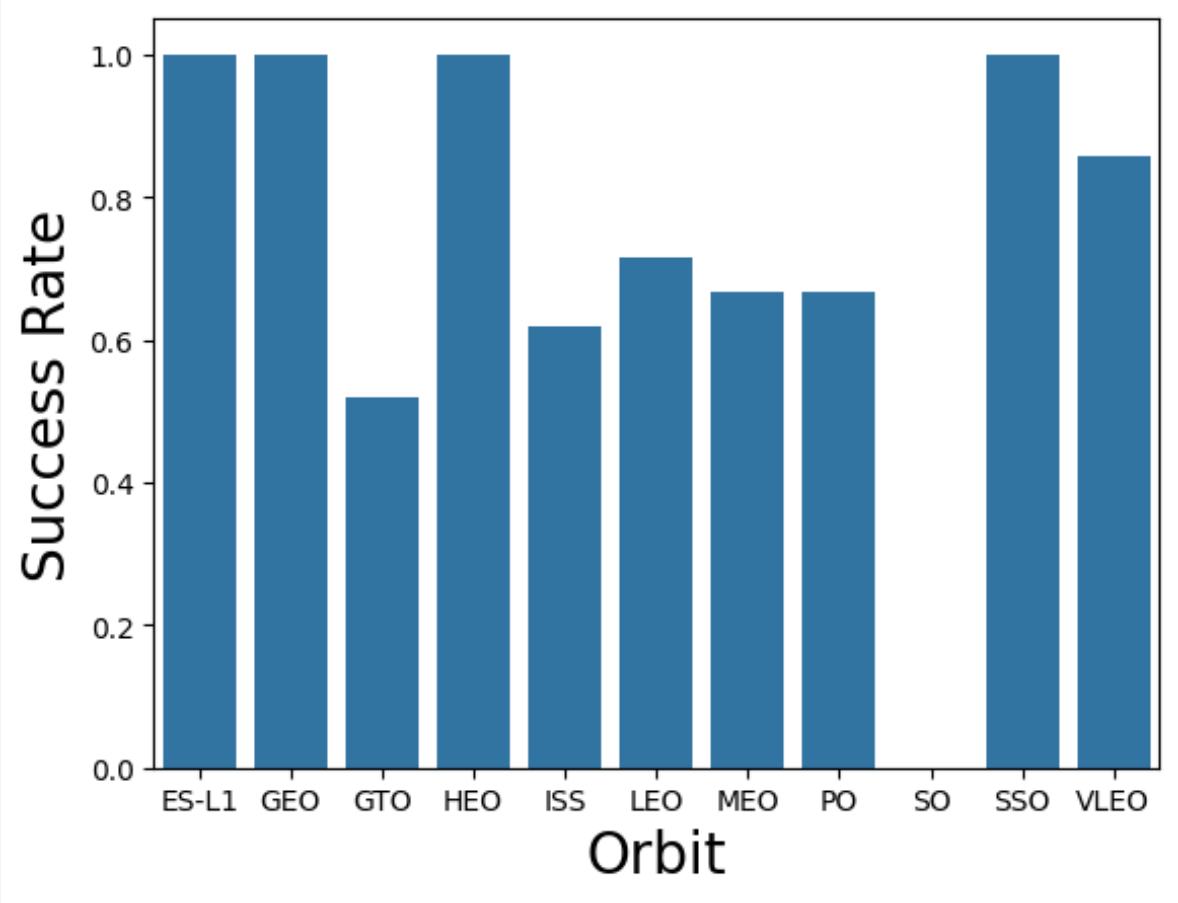
Payload vs. Launch Site



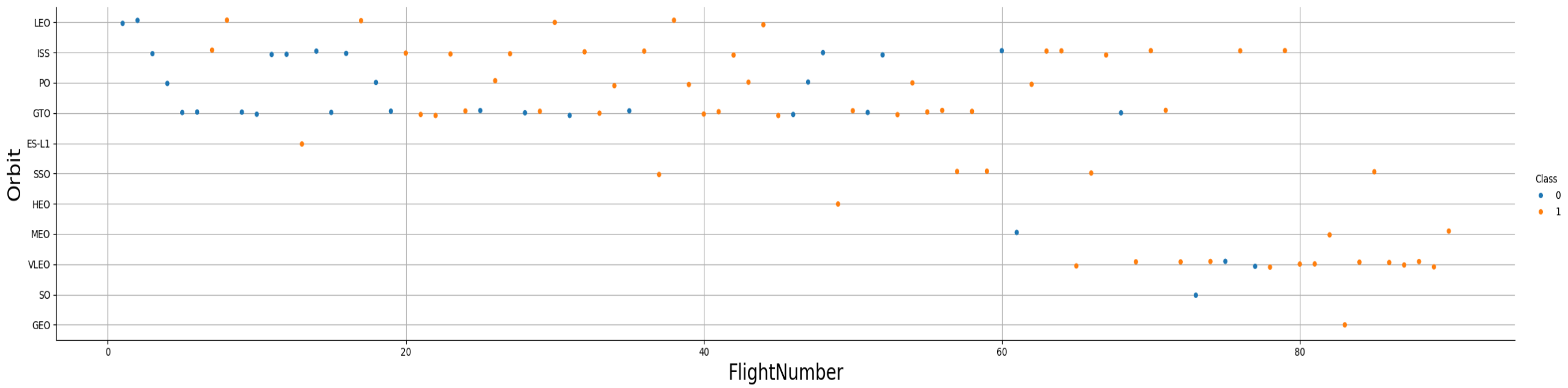
- It can be seen that the site VAFB SLC 4E does not handle heavy loads, in other sites the successful rate seems to be better for heavier loads.

Success Rate vs. Orbit Type

- Orbits ES-L1, GEO, HEO y SSO has a 100% successful rate, while orbit GTO is more challenging.
- The SO orbit only has one launch, which is why it is not taken into account in this analysis.

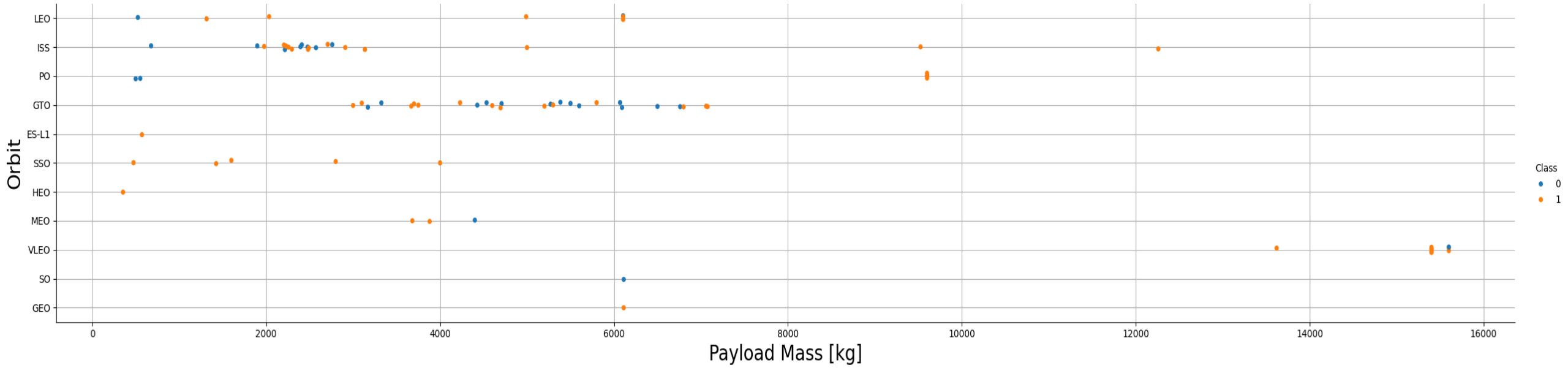


Flight Number vs. Orbit Type



- It can be seen that as increases de flight number, increases the rate of success in all orbits, it indicates the increment into technical learning curve.
- The LEO orbits shows a concentration of early success, hile GTO exhibits greather dispersion and volume of attempts over time.

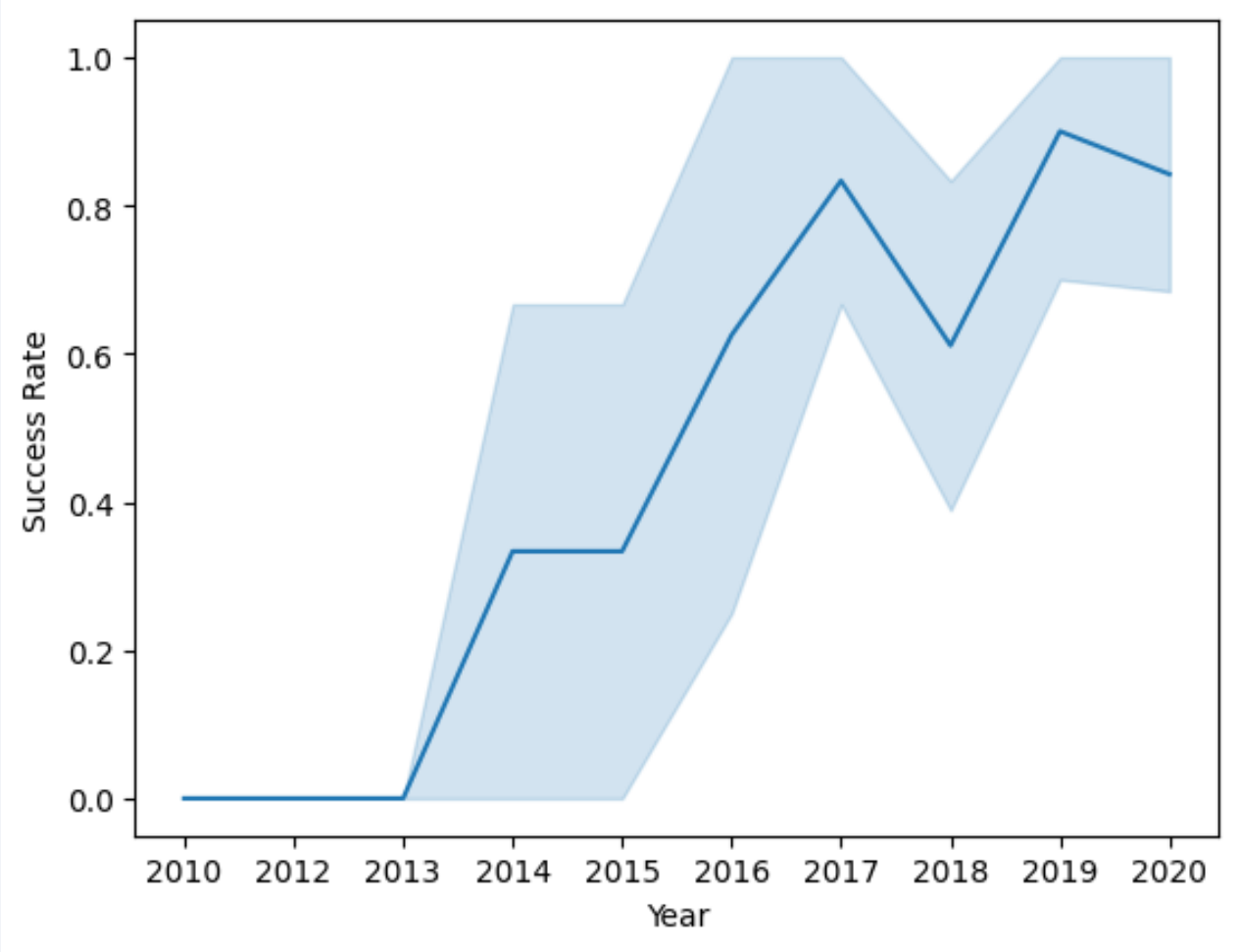
Payload vs. Orbit Type



- Polar orbits (PO and SSO) maintains lower an medium loads, while LEO supports greater weights.
- No failed landing seen with the highest loads in LEO orbits, suggesting that high capacity missions use boosters with more reliability.

Launch Success Yearly Trend

- The line shows a clear upward trend since 2013, demonstrating that reuse technology has grow significantly over time.



All Launch Site Names

- Names of the unique launch sites:
  - CCAFS LC-40: Cape Canaveral Space Launch Complex 40
  - VAFB SLC-4E: Vandenberg Space Launch Complex 4
  - KSC LC-39A: Kennedy Space Center Launch Complex 39A

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- The image shows 5 records where launch sites begin with `CCA`

Total Payload Mass

- The image shows the total payload carried by boosters from NASA

SUM(PAYLOAD_MASS__KG_)
45596

Average Payload Mass by F9 v1.1

- The image shows the average payload mass carried by booster version F9 v1.1

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

First Successful Ground Landing Date

- The image shows the date when the first successful landing outcome in ground pad was achieved.

MIN(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- The image shows names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- The image shows the total number of successful and failure mission outcomes

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- The image shows all the booster_versions that have carried the maximum payload mass

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Records with names of the months, the results of failed landings on the unmanned ship, the booster versions and the launch site for the months of the year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

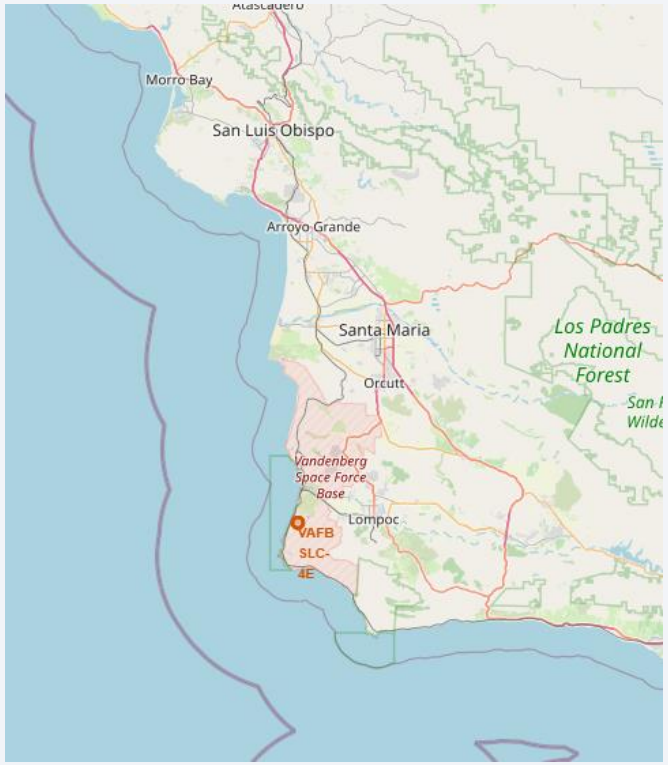
- The image shows the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

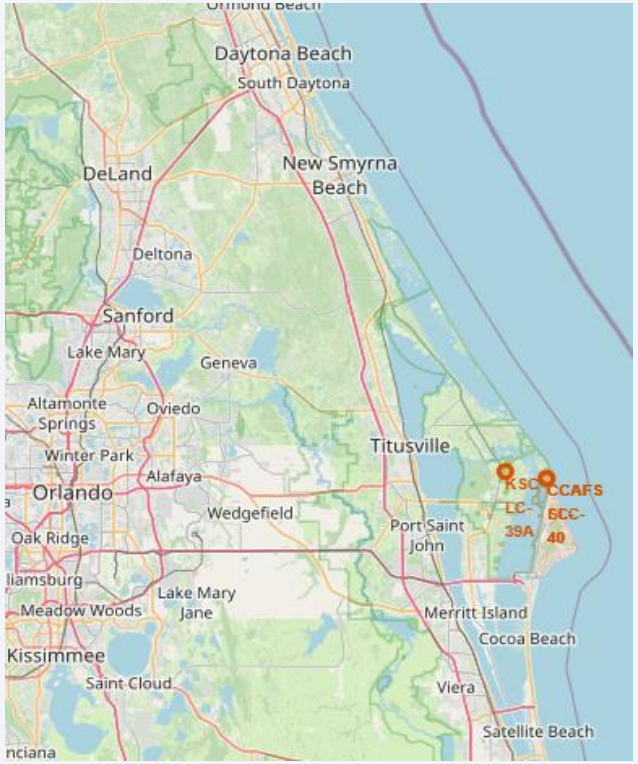
Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

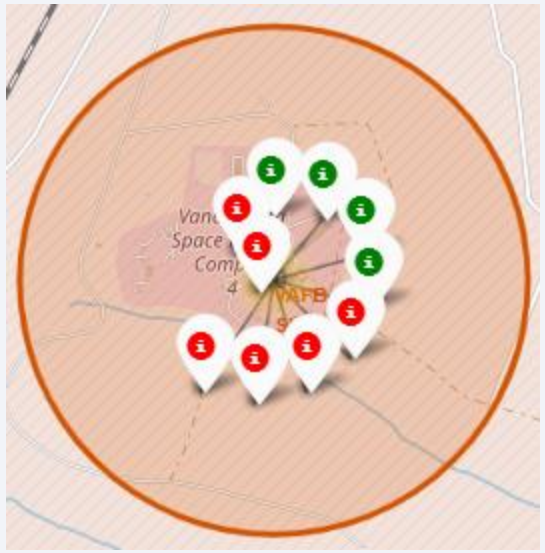
Launch Sites Locations

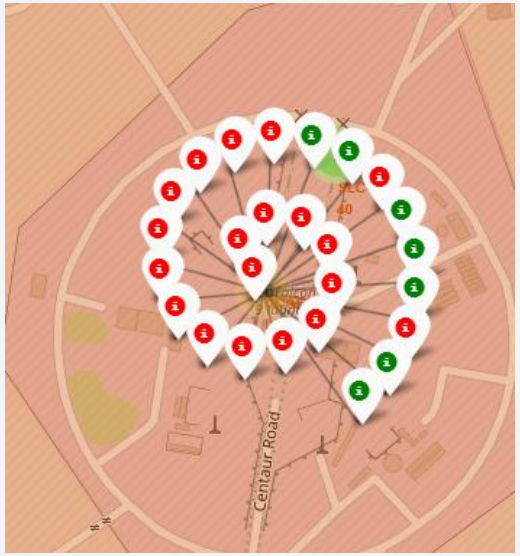


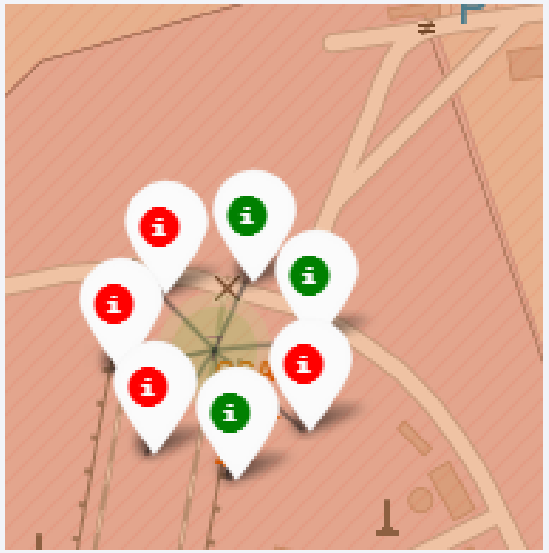


- The launch sites are strategically located on the coasts to facilitate trajectories over the ocean.

Success and Failed Landings by Launch Site

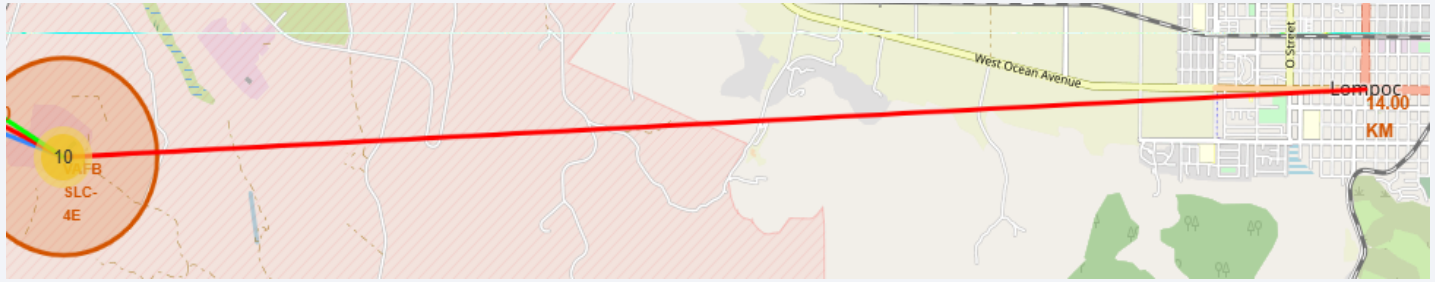






- Each image represent a launch site, in order: VAFB SLC-4E, KSC LC-39A, CCAFS LC-40, CCAFS SLC-40. Last two in the same property.
- Green markers represent success landings and red markers failed.
- Can see sites with more failure rate and sites with more reliability as KSC LC-39A.

Proximity and Distances Analysis



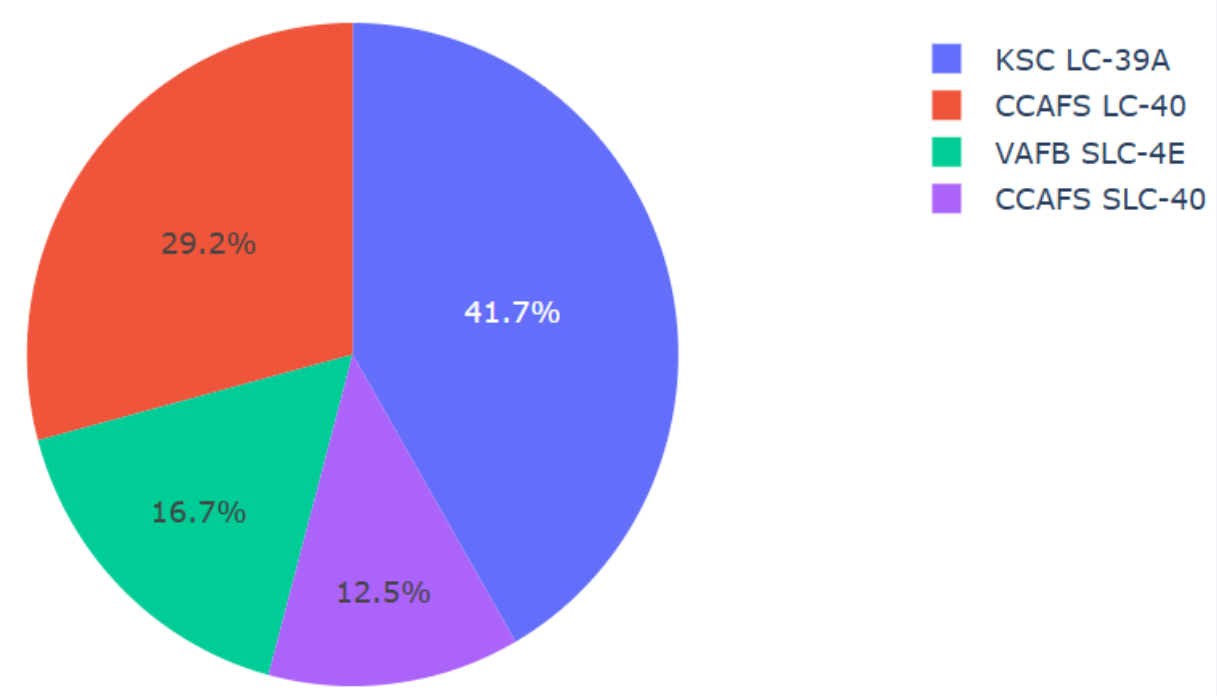
- In the image on the left, the lines indicate the distance from the launch site 'name' to the nearest coast, railways and road; on the right, the distance to the nearest city.
- Launch sites are intentionally located: near to infrastructure (railways, highways) to facilitate de logistical transport; near to coast an far from cities to minimize risks to populated areas during ascent or failure.

Section 4

Build a Dashboard with Plotly Dash

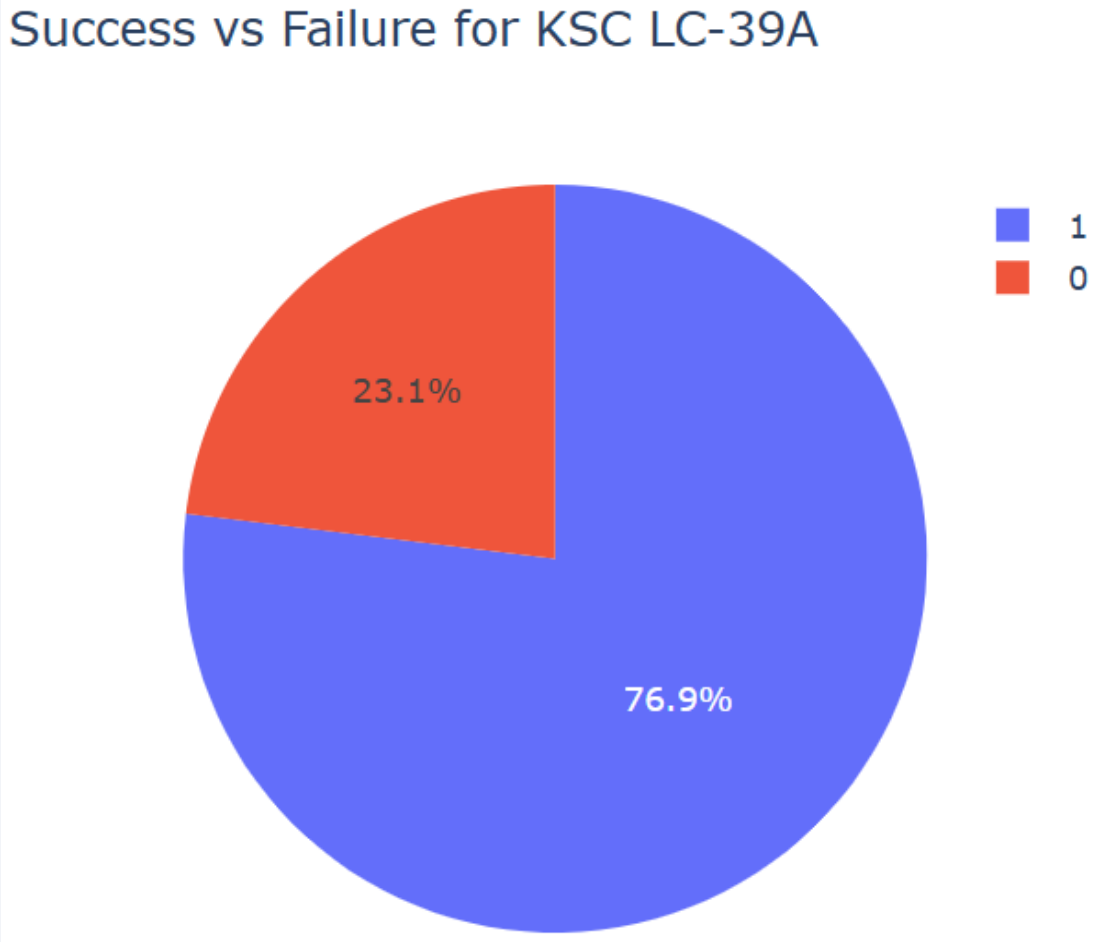
Launch success count for all sites

- It can be seen that KSC LC-39A has the most successful launches.

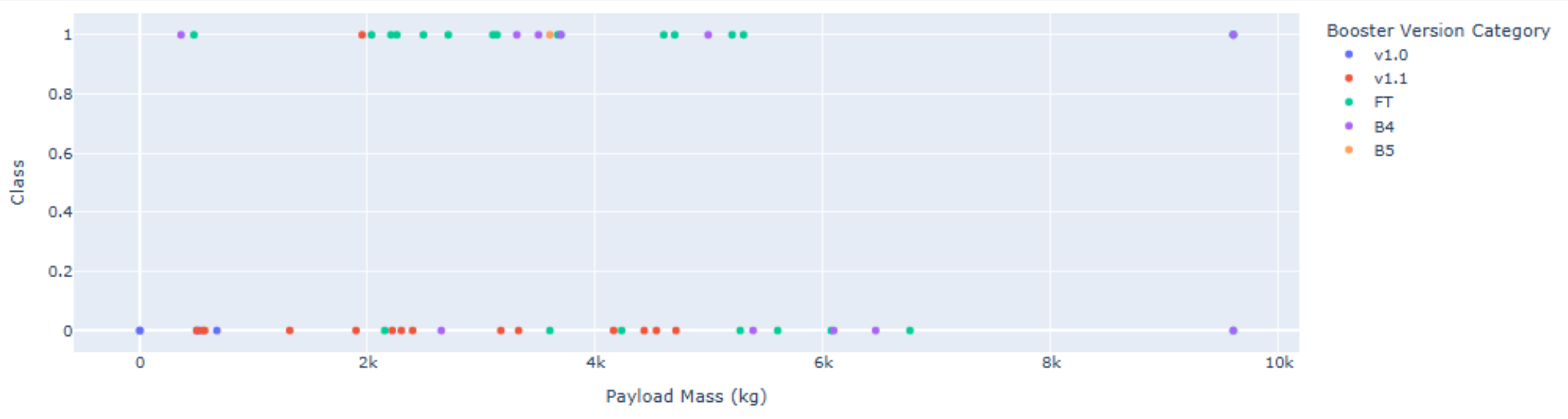


Launch site with highest launch success ratio

- It can be seen that the launch site with highest launch ratio success is the same site with more successful launches.



Correlation between payload and success for all sites



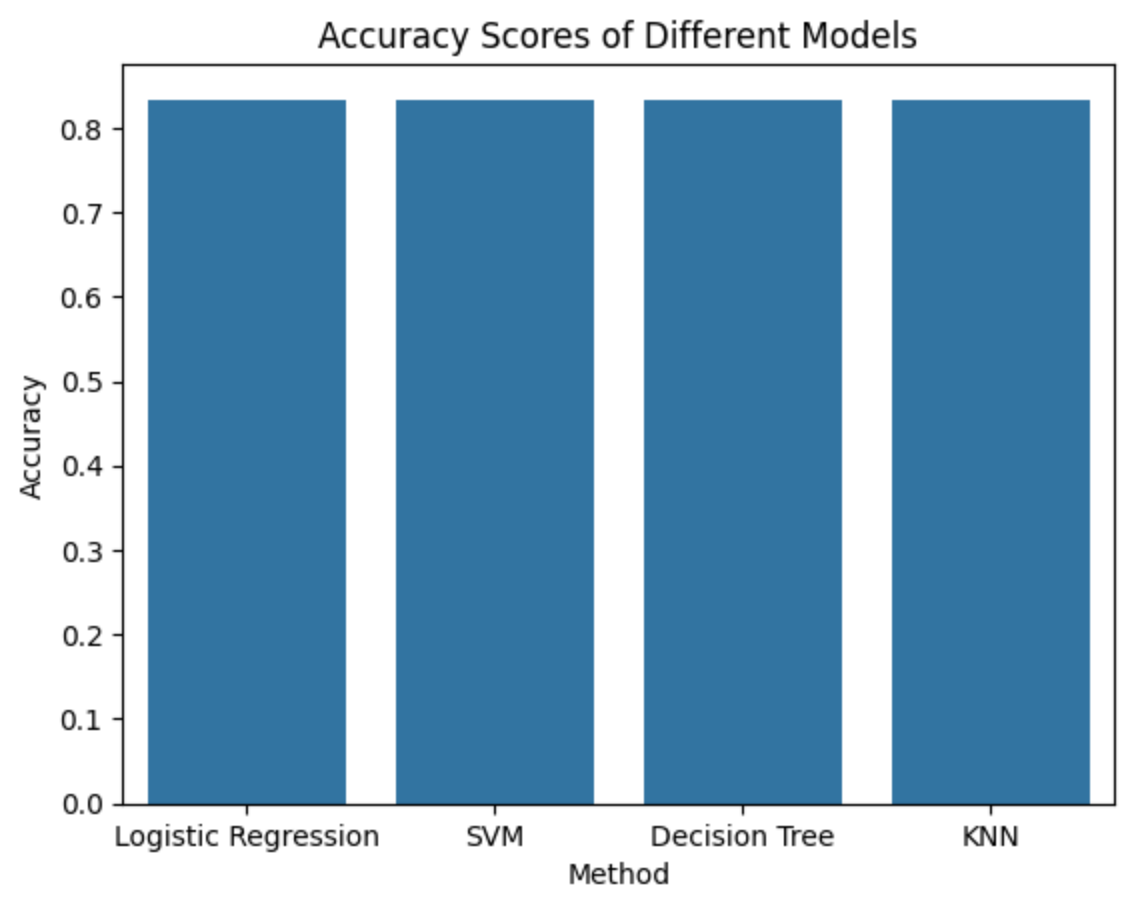
- The booster version B5 has best success rate while FT or v1.1 (older versions) has worse successful rate.
- It can be seen high successful rate between 2000kg and 5000 kg payload.
- In higher loads the most success are from newest booster versions.

Section 5

Predictive Analysis (Classification)

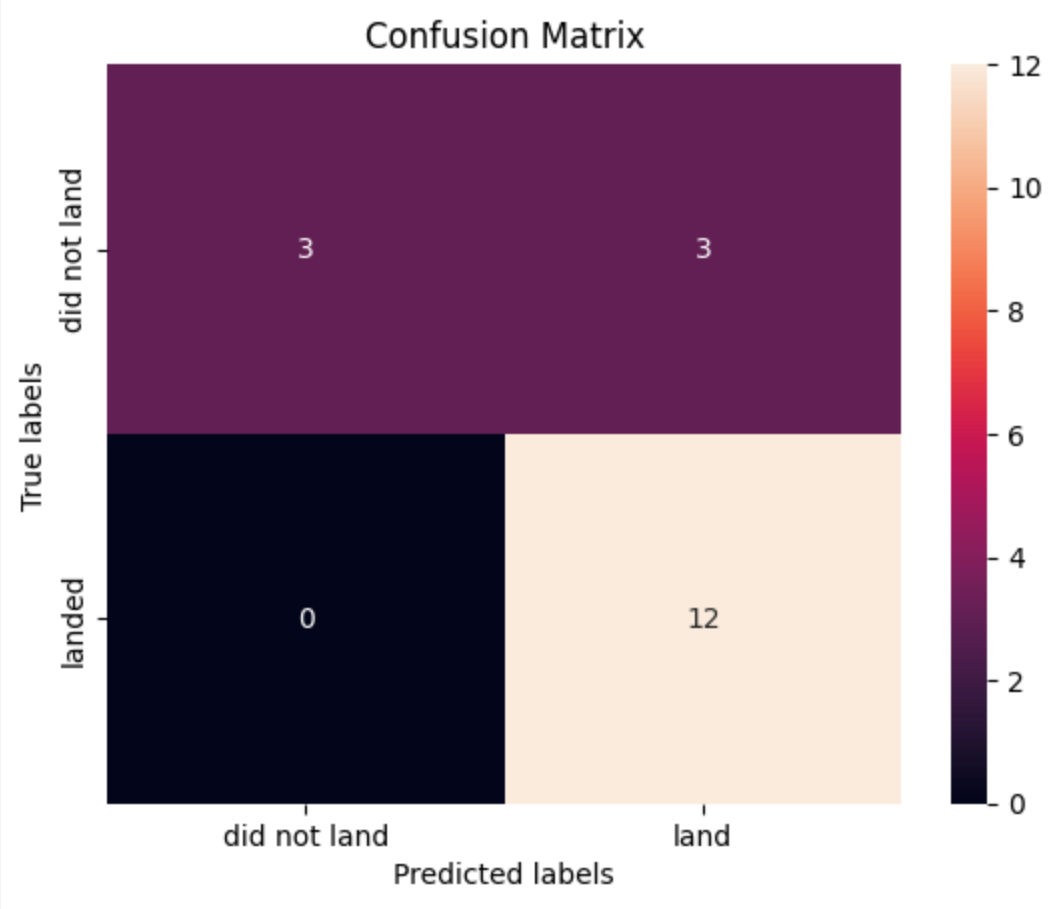
Classification Accuracy

It can be seen that all models achieved identical accuracy. In this case, one could opt for the model with the best interpretability such as Logistic Regression or Decision Tree.



Confusion Matrix

- All the models has identical confusion matrix. It probably indicates that the best performance for the actual dataset was achieved.



Conclusions

- Technological maturity is measurable: historical analysis confirms that SpaceX has passed the experimentation stage. Reusing is not a technical risk, but an operative advantage.
- Success factors identified: proximity to coasts is not only for security, it is a logistical necessity for the recovery of drone ships; was identified that polar orbits (PO and SSO) maintains lower an medium loads, while LEO supports greater weights.
- Power of prediction: we achieved a model with an accuracy of 83.3%. This can predict the success of a mission before the launch. The tie between the models suggest that we have captured the essence of the avaible data. This tool allows to evaluate the financial risk of a mission.

Appendix

- Data Glossary:
  - GridFins: Boolean that indicates if mesh fins was used.
  - Reused: indicates if booster was used before.
- Hyperparameters configuration:
  - Logistic Regression: {'C': [0.01, 0.1, 1], 'penalty': ['l2'], 'solver': ['lbfgs']}
  - KNN: parameters = {'n_neighbors': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10], 'algorithm': ['auto', 'ball_tree', 'kd_tree','brute'], 'p': [1,2]}
  - SVM: {'kernel': ['linear', 'rbf', 'poly'], 'C': [0.5, 1, 1.5]}
  - Decision Tree: {'criterion': ['gini', 'entropy'], 'max_depth': [4, 6, 8]}

Appendix

	FlightNumber	PayloadMass	Flights	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	Orbit_GTO	Orbit_HEO	Orbit_ISS	...
0	1.0	6104.959412	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...
1	2.0	525.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...
2	3.0	677.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	...
3	4.0	500.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...
4	5.0	3170.000000	1.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...

5 rows x 83 columns

...	Serial_B1058	Serial_B1059	Serial_B1060	Serial_B1062	GridFins_False	GridFins_True	Reused_False	Reused_True	Legs_False	Legs_True
...	0.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
...	0.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
...	0.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
...	0.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0
...	0.0	0.0	0.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0

- Head of final DataFrame used to train models (without target variable 'Class')

Thank you!